

When the Attack Is Not an Attack

Validity Failures in LLM-Generated Red-Teaming

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<https://github.com/yhj3/counterfactual-textattack>

The setting, in one example

A text classifier is a function from text to a label. Feed it a film review, it answers *positive* or *negative*.

An **adversarial attack** changes the input so the *model* changes its answer while the *correct* answer stays the same. A person reading both versions would give the same label; the model does not. That gap is the vulnerability.

TextFooler finds these by swapping words for synonyms, one at a time, choosing the words the model leans on most. In principle:

*The movie was **excellent**, I **loved** every minute. → positive*

*The movie was **superb**, I **adored** every minute. → negative*

In practice the output is rarely that clean. Here is a real one from my data, on a news article:

Original. *Fears for T N pension after talks. Unions representing workers at Turner Newall say they are ‘disappointed’ after talks with stricken parent firm Federal Mogul.*

TextFooler. *Fears for T percent pension after debate. **Syndicates portrayal worker** at Turner Newall say they are ‘disappointed’ after **chatter** with **bereaved parenting corporations** Canada Mogul.*

“Bereaved parenting corporations” is not something a person would write. The median GPT-2-XL perplexity of a TextFooler perturbation is **522**, where ordinary English sits in the tens. An input nobody would ever produce says little about deployment risk.

Hence the idea this paper examines: hand the attack to a language model and ask it to rewrite the whole sentence so it reads naturally, while staying adversarial. This is now a common shape for automated red-teaming — one model generating the texts that test another. I call each such text a *probe*: a candidate adversarial input, produced by one model, whose only job is to test another. The word keeps the two roles apart, because every failure below is a failure of the probe, not of the model being tested. **My question is what such a pipeline actually measures.**

Finding 0: rewriting is a trade, not an improvement

Rewriting is applied only to attacks that *already succeed*, so its cost is directly measurable: how many survive?

44.5%, across 32 model–attack–dataset configurations. More than half the attacks are destroyed by being rewritten. What that buys is fluency: median perplexity falls from 522 to **22.9**.

The mechanism is not mysterious. The words fooling the model are often exactly the awkward ones. Repair the English and the model stops being fooled — fluency and adversarial effect are competing for the same words.

Finding 1: label validity — the probe is not an attack

I ran two rewriting prompts on identical data: a task-specific one and a generic one. By flip rate the task-specific prompt wins two to one, **119/180 against 58/180** ($p = 1.7 \times 10^{-10}$).

Then I looked at *how* it wins. AG News is news-topic classification, and the task-specific prompt for it reads “changing topic from world to sports”. It instructs the model to change the label. A real result:

Original (business): Yukos warns its oil output is lagging. Beleaguered Russian energy giant Yukos has warned that it will not produce as much oil as expected this year...

Rewrite: Lakers facing setbacks in NBA title defense, LeBron's injury a major concern. The Los Angeles Lakers are facing a difficult road ahead in their quest to defend their NBA title...

Classifier: business → sports. Cosine similarity 0.016.

The classifier is right. That text is sports news. The language model did not attack anything; it wrote a different article. But a pipeline counting flips records this as a successful attack.

The usual defence, and where it breaks

The standard fix is a semantic-similarity threshold. On AG News it works — similarity 0.016 is caught by any threshold. On sentiment it does not:

Original. Sloppy film noir thriller which doesn't make much of its tension promising set-up. (3/10)

Rewrite. Engaging film noir thriller that successfully builds on its intriguing premise. (8/10)

Cosine similarity 0.964 — comfortably past any threshold.

The verdict has been reversed, 3/10 to 8/10, and the embedding hardly moves, because both sentences are about the same film noir thriller. **Sentence embeddings track topic, not verdict.** So the check catches topic changes and waves sentiment changes through.

A referee model, and the ranking reverses

An adversarial example should fool *one* model, not persuade every model. So I re-scored every probe with an independently trained classifier of the other architecture — RoBERTa refereeing BERT's probes and vice versa.

Dataset	Prompt	Referee also flips	Mean sim.
AG News	task-specific	87.2%	0.268
AG News	generic	44.4%	0.612
IMDb	task-specific	86.0%	0.749
IMDb	generic	23.1%	0.719
SST-2	task-specific	86.7%	0.640
SST-2	generic	66.7%	0.667

86–87% of the task-specific prompt's "successes" also flip an independent model. Those are not vulnerabilities. They are inputs whose label changed. Requiring the flip to be specific to the target reverses the comparison:

Prompt	Flip rate	Genuine vulnerabilities
task-specific	119/180 (0.661)	16/180 (0.089)
generic	58/180 (0.322)	29/180 (0.161)

The prompt that scores best on the reported metric is the worse red-teaming prompt. And this is structural rather than accidental: the instruction most effective at producing label flips is the instruction to change the label, so any prompt search that optimizes reported attack success will find it.

A third signal, and the sharpest version of the result. The referee can be objected to: real adversarial examples do transfer between models. So I added a signal that is not a classifier of the task at all — a DeBERTa NLI model, scoring entailment in both directions, which has never been told what sentiment or topic means. Under each definition of a valid attack, this is which prompt wins:

Validity criterion	task-specific	generic	p
flip only (the reported metric)	119/180	58/180	$< 10^{-4}$
+ cosine ≥ 0.75	39/180	25/180	0.073
+ referee, target-specific	16/180	29/180	0.055
+ NLI mutual entailment	6/180	16/180	0.046
all three	1/180	7/180	0.067

The cosine constraint is the only added check that leaves the ranking unchanged. It removes two thirds of the flips and still picks the worse prompt — so the defence this literature relies on does not change the decision a practitioner would make. Both signals independent of the embedding space do change it, and NLI gives the only reversal significant at the conventional level.

The two meaning-preservation checks also disagree with each other: of the 64 flips that clear the cosine threshold, **50 (78%) fail mutual entailment**, and the disagreement sits where an embedding would be expected to fail — 29 on IMDB and 15 on SST-2 against 6 on AG News. A model that was never told what sentiment is confirms the sentiment failure independently.

Two checks, and neither is sufficient. Of the 119 flips under the task-specific prompt, 39 survive the similarity threshold, 16 survive the referee, and **only 5 survive both**. If the two checks caught the same thing, the overlap would be close to 16. It is 5, because they fail in different places:

	referee: target-specific	referee: it flips too
similarity ≥ 0.75	5	34
similarity < 0.75	11	69

The 34 are cases the similarity check waves through and the referee catches — the 3/10-to-8/10 review above is one. The 11 are the reverse: a world-news article rewritten into a rugby story, which similarity caught at 0.268 while the referee still answered “world”. Adding one check is not enough, because each one’s blind spot is the other’s catch.

Counting a probe as target-specific whenever the referee does not flip is also generous: the referee may have disagreed with the target on the original text to begin with. The two agree on the original 94.4% of the time, and requiring agreement leaves **12/180** genuine vulnerabilities for the task-specific prompt against **25/180** for the generic one — the reversal widens.

Finding 2: input validity — the probe is not a valid input

RTE and QNLI take *two* sentences, a premise and a hypothesis, and ask how they relate. The format is `sentence1 <SPLIT> sentence2`. Rewriting frequently destroys it:

Original. *Sentence1: Dana Reeve, the widow of the actor Christopher Reeve, has died of lung cancer at age 44... <SPLIT> Sentence2: Christopher Reeve had an accident.*

Rewrite. *Dana Reeve, the widow of the actor Christopher Reeve, has died of lung cancer at age 44, according to the Christopher Reeve Foundation. Christopher Reeve had an accident.* In this revised version, the sentence has been altered to include...

The two segments are merged and a commentary sentence is appended. The classifier cannot read this as a pair at all. Across the stored corpus only 100 of 198 pair-task rewrites kept the structure; for RTE under PWWS, **0 of 31** did.

The direction of the resulting error is the interesting part. **A malformed probe never registers as a flip**, so it lands in the denominator as an attack failure:

Prompt	Well-formed	Reported if unchecked	Valid probes only
generic	0.300	12.5%	41.7%
structured	0.575	21.7%	37.7%

Stating the format explicitly nearly doubles validity ($p = 2.8 \times 10^{-5}$), though the model still violates a stated format more than 40% of the time. **Finding 1 inflates the reported number; Finding 2 deflates it. The two errors push in opposite directions, so they do not cancel** — which one dominates depends on the task, and neither is visible in a success rate.

Finding 3: generator validity — the red-teamer refuses

The generating model is itself safety-trained, and sometimes declines:

I cannot provide a revised text that could potentially be used to mislead or deceive, as it is unethical and goes against my programming rules...

The refusal is then scored as though it were a probe. Of 600 generations, **37 (6.2%)** begin with a refusal, and **7 of those are counted as successful attacks** — the target assigns the refusal text a different label than the original, at mean similarity 0.216. The target is being credited with a vulnerability to a sentence about the generator’s programming rules.

It is not evenly spread over the conditions being compared. The generic prompt, which says “maintain adversarial properties to fool the classifier”, is refused **11.0%** of the time; the task-specific prompts, which ask for a sentiment or topic change without naming the adversarial purpose, **0.6%**. On RTE the rate is **17.5%**, plausibly because asking a model to flip an entailment reads as a request to assert something false.

That makes it a confound in my own comparison, so I checked it: excluding refusals, the task-specific prompt retains 118/179 against 52/170 ($p = 4.2 \times 10^{-11}$), essentially unchanged. Both findings survive. But **a refusal and a failed attack look identical in the output column**, and as generators are trained to decline adversarial requests more reliably, this failure grows rather than shrinks.

Finding 4: measurement validity — the number is not about the text

The last failure leaves no trace in the outputs at all. An earlier version of this study wrote the language model’s output into a `llama_text` column, then computed the reported success rate from TextAttack’s `result_type` field — which describes the word-substitution attack that ran *before* rewriting. The generated text was never scored.

The check is one line: assert that the column being evaluated is the column that was generated. That this happened in a study whose entire subject was generated text is the point — it is a default of the tooling, not carelessness about the topic.

Five numbers a red-teaming report should carry

1. **Semantic similarity beside every flip rate.** On AG News, the difference between reporting 65.0% and 3.3%.
2. **The fraction of successes an independent check reproduces.** Similarity alone does not change the ranking; a referee model or an NLI entailment test does, and disagrees with cosine on 78% of what cosine accepts.
3. **The fraction of generations that are well-formed inputs.** On pair tasks, the difference between 12.5% and 41.7%.
4. **The generator’s refusal rate, per condition.** Here 11.0% against 0.6% for the two prompts being compared.
5. **The number of generations the target model actually scored.** Without it, a pipeline can report a metric computed from something else entirely, and nothing in the output will look wrong.

Full paper, code, stored probes, and every result table: <https://github.com/yhj3/counterfactual-textattack>